

(b) Discuss different types of primary interatomic bonds.

There are three main types of primary interatomic bonds that exist between atoms in a material: covalent, ionic and metallic.

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1. **Ionic bond**: An ionic bond is formed when one or more electrons are transferred from one atom to another. This results in positively charged cations and negatively charged anions that are attracted to each other by electrostatic forces. Ionic bonds are typically found in compounds composed of a metal and a non-metal. Ionic bond presents in  $\text{NaCl}$ ,  $\text{MgCl}$ ,  $\text{CsCl}$ ,  $\text{AlCl}_3$  etc.

2. **Co-valent bond**: Covalent bonds form when atoms share one or more pairs of electrons to achieve a stable electron configuration. This type of bond typically occurs between non-metal elements. Covalent bonds can be polar or non-polar depending on the electro-negativity difference between the atoms. Examples of covalent compounds are  $\text{H}_2\text{O}$ ,  $\text{CO}_2$  etc.

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1. Ionic bond: An ionic bond is formed when one or more electrons are transferred from one atom to another. This results in positively charged cations and negatively charged anions that are attracted to each other by electrostatic forces. Ionic bonds are typically found in compounds composed of a metal and a non-metal. Ionic bond presents in NaCl, MgCl, CsCl, AlCl<sub>3</sub> etc.

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2. Co-valent bond: Covalent bonds form when atoms share one or more pairs of electrons to achieve a stable electron configuration. This type of bond typically occurs between non-metal elements. Covalent bonds can be polar or non-polar depending on the electro-negativity difference between the atoms. Examples of covalent compounds are H<sub>2</sub>O, CO<sub>2</sub> etc.

3. Metallic bond: Metallic bond occurs between metal atoms and are characterized by the pooling of valence electrons among a sea of atoms. This allows the metal atoms to share electrons, leading to a cohesive, strong bond. Metallic bonds are responsible for the high strength and ductility of metals. Examples of metals with metallic bonds include iron, copper and gold.

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(c) Calculate the force of attraction between a  $\text{Ca}^{2+}$  and an  $\text{O}^{2-}$  ion the centers of which are separated by a distance of 1.25 nm

We know,

$$F = \frac{k q_1 q_2}{r^2}$$

$$= \frac{9 \times 10^9 \times 2 \times -2}{(1.25 \times 10^{-9})^2}$$

where,

$F$  = Force of attraction

$r$  = distance between the centers of two atoms

$q_1, q_2$  = charges of two ions

From Callister

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{|z_1| |z_2| e \times e}{r^2}$$

$$r = 1.25 \text{ nm} = 1.25 \times 10^{-9} \text{ m}$$

$$K = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

$$q_1 = +2e \text{ (the charge of } \text{Ca}^{2+})$$

$$q_2 = -2e \text{ (the charge of } \text{O}^{2-})$$



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Ques - 2020

(a) What are the different types of primary bonds in solids? Schematically explain the metallic bonding.

→ 2021 ques 2(b)

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(b) Discuss various types of Vander Waal's forces.

Vander waal's forces are a type of forces intermolecular force that occurs between molecules or atoms. These forces arise due to the fluctuations in the electron distribution around a molecule which creates temporary dipoles. Vander waal's forces are relatively weak compared to other types of chemical bonds. There are three types of vander waal's forces.

1. London dispersion forces: London dispersion forces are the weakest type of vander waal's force. They arise from the instantaneous dipoles can induce similar dipoles in nearby molecules, creating a weak attractive force between them. London dispersion forces are present in all molecules and increase with the size of the molecules and the number of electrons it contains.

2. Dipole-dipole forces: Dipole-dipole forces arise when molecules have a permanent dipole moment. which occurs when there is an electronegativity difference between the atoms in a molecule. These forces are stronger than London dispersion forces and act between polar molecules. Dipole-dipole forces cause the molecules to align with each other, with the positive end of one molecule attracting the negative end of another.

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3. Hydrogen bond: Hydrogen bond is a type of dipole-dipole force that occurs when a hydrogen atom is bonded to a highly electronegative atom such as oxygen, nitrogen or fluorine. The hydrogen atom in this bond carries a partial positive charge, while the electronegative atom carries a partial negative charge. The hydrogen atom can form a weak bond with the electronegative atom of another molecule, creating a strong intermolecular force.

(c) Write down the properties of ionic bond and covalent bond in solids.

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### Properties of ionic bond

1. High melting and boiling point
2. Solubility in polar solvents
3. Brittle
4. High conductivity
5. Regular crystal structure
6. Strong electrostatic force.

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### Properties of covalent bond

1. Low melting and boiling point
2. Insoluble in polar solvent/Solubility in non-pol
3. Malleable and ductile
4. Non-conductivity
5. Specific arrangement in molecular structure
6. Both polar and non-polar bonds.



## Properties of metallic bond

1. High melting and boiling point

2. Malleable and ductile

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3. Good conductors of electricity

4. Lustre

5. Solid at room temperature

6. Alloy formation.

Ques - 2019

2. (a) Cite two important quantum-mechanical concepts associated with the Bohr model of the atom.

1. that electrons are particles moving in discrete orbitals and 2. electron energy is quantized into shells.

(b) Give the significance of each of the four quantum numbers.

The four quantum numbers are:

1. Principle quantum number ( $n$ ): The principle quantum number describes the energy level of an electron in an atom. It determines the size and energy of the electron's orbital. The higher the value of  $n$ , the longer and more energetic the orbital is.

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2. Azimuthal quantum number ( $l$ ): The azimuthal quantum number determines the shape of an electron's orbital. It can have any integer value from 0 to  $n-1$ , where  $n$  is the principal quantum number. The value of  $l$  determines the subshell in which an electron resides.

3. Magnetic quantum number ( $m$ ): The magnetic quantum number describes the orientation of an orbital in three-dimensional space. It can have any integer value between  $-l$  to  $+l$ , including 0. The value of  $m$  determines the specific orbital within a subshell that an electron occupies.

4. Spin quantum number ( $s$ ): The spin quantum number describes the intrinsic spin of an electron which is a fundamental property of particles. It can have a value of either  $+\frac{1}{2}$  or  $-\frac{1}{2}$ , representing the two possible spin states of an electron.



(d) Metals are good conductors of both electricity and heat whereas polymers and ceramics are not.

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The ability of a material to conduct electricity and heat is related to its electronic and molecular structure. Metals have a unique arrangement of atoms that allows them to conduct electricity and heat efficiently. In metals, the outer most electrons of each atom are not tightly bound to their respective nuclei and can move freely throughout the metal. The movements of these free electrons are responsible for the high electrical conductivity of metals. Additionally, the arrangement of atoms in metals allows for the efficient transfer of thermal energy through the lattice structure, making them good conductors of heat.

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On the other hand, polymers and ceramics have a different structure. Polymers are made of up of long chains while ceramics are typically made up of ionic or covalent bonded compounds. In both cases, the electrons are not as free to move as they're in metals. In addition, the molecular or ionic structure is less efficient at transferring thermal energy, leading to poor thermal conductivity.